

During the “pioneering phase”, construction relied on the expertise of engineers – civil and military. Before 1870, many icons of the British Railways were brought in as consultants. Isambard Kingdom Brunel, a prolific engineer of his time, consulted for the Eastern Bengal Railway, and outlined proposals for the Sealdah Terminus station in Calcutta. British contractors, such as Victorian railway builder Thomas Brassey, executed the building of lines, employing vast numbers of Indian labourers. There were some British Indian contractors and the Indian Jamsetjee Dorabjee, who secured four contracts.

Skilled and experienced workers came to India for construction. In 1861, according to Canadian historian Ian J Kerr, there were about 500 Britishers overseeing the building of the railway lines across India, sharing their knowledge with local workers. Technology transfer, wrote Kerr, was a core element of construction and operation.

Material matters

All know-how was British, whether it was rulebooks, procedures, or equipment. All signalling equipment, machinery, wheel units for carriages, and rails were imported from Britain. By the end of 1863, noted Kerr, 3,571 ships carrying nearly three million tonnes of railway materials had made their way to India. By the end of 1869, some 1,045 British locomotives were in operation. This was also the time when workshops were set up – not to construct locomotives, but for minor repairs. The first workshop was set up in 1854 in Bombay followed by one in Calcutta. The iconic [Jamalpur Workshop](#) was constructed in 1862.

Back-breaking work

One of the least technical, yet most labour-intensive processes was building rail beds. This included extensive earthwork and building embankments to create ground levels and set rails. There were several instances where masonry and brickwork were involved in order to protect the tracks against floods, as in Bengal. Most of the workers were unskilled and hired from nearby villages. For heavy construction in remote areas, labourers were even hired from far-flung places.